

Nitya Patel

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SUMMARY

Software engineer with 2 years of professional experience building and owning production Python systems across quantitative trading infrastructure, applied ML, and AR/VR. Consistently delivers high-impact internal tooling with measurable outcomes.

EDUCATION

IIIT Vadodara

B.Tech in Computer Science | GPA: 8.61/10.0

Gandhinagar, Gujarat

2020 – 2024

Shri Agrasen Vidyalaya

Higher Secondary Certificate | Score: 84.6%

Indore, Madhya Pradesh

2019

EXPERIENCE

Graviton Research Capital

Software Developer – Tech Tools

Gandhinagar / Gurugram

June 2024 – Present

- Develop and maintain daily automations and exchange-specific workflows across 4 exchanges.
- Built and owned the `config_converter` module, which runs for 1000s of runs daily, saving developer effort and time.
- Designed a usage-based borrow allocation algorithm used across ~30 strategies.
- Create internal tooling to eliminate repetitive tasks and improve team productivity.

VizExperts

Software Development Intern

Gurugram

January 2024 – May 2024

- Designed and built an AI-based live drift correction module from scratch using Python, OpenCV, and PyTorch, targeting iPad AR workflows without multi-camera hardware support.
- Achieved a ~10–15× reduction in drift accumulation rate compared to baseline.
- Assisted with bug resolution in an AR/VR application built on Unreal Engine using C++ and Blueprint scripting.

MathWorks

EDG Intern

Bangalore

May 2023 – July 2023

- Added support for FMU 3.0 in the FMU Import Block component in Simulink.

iHub-Data, IIIT Hyderabad

Machine Learning Intern

Hyderabad

May 2022 – July 2022

- Contributed to a Dataset Foundation project to create a platform for ML/AI researchers.
- Collaborated with a team of 10 to design the project architecture and establish an efficient delivery pipeline.
- Engineered a high-performing, multithreaded download manager for datasets up to 100 GBs.

PROJECTS

Algorithmic Trading Framework | Python, asyncio, Parquet

- Architected a single-process trading infrastructure with a central event bus — components are fully decoupled, and new exchanges or strategies are each a single-file addition.
- Implemented a live CoinDCX exchange adapter for OHLCV candle streaming and defined a BaseExchange interface to support multi-exchange operation without architectural changes.

Ornithological Sound Identifier | Python, PyTorch

- Implemented a multilayered neural network that identifies birds from soundscapes.

Todoist Command Line Utility | Python

- Developed a command line utility for Todoist using the Todoist REST API.

TECHNICAL SKILLS

Languages: Python, C, C++, Java, JavaScript, Matlab, SQL, Bash

Tools & Frameworks: Git, Django, ReactJS, LaTeX, Firebase, PyTorch

Operating Systems: Windows, Linux

ACHIEVEMENTS

Giftabled Hackathon: 2nd rank among 100 participants — created a facial recognition app embedded on an attachable pin, focused on helping visually impaired people.

CTF: 3rd rank among 100 participants — Capture the Flag; solved sequential puzzles involving cryptography, computer networks, web technology, and maths.

Tech Hunt: 1st rank among 150 participants — non-sequential puzzle competition; most puzzles solved wins.